

Eshan Raul

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Research Interests

Astrobiology & habitability

Exoplanetary atmospheres/dynamics

Astrochemistry

Education

University of Wisconsin–Madison

May 2026–May 2029

Ph.D. in Astronomy

Advisor: Thomas Beatty

University of Wisconsin–Madison

August 2024–May 2026

M.S. in Astronomy

Advisors: Coco Zhang, Juliette Becker

University of Michigan, Ann Arbor

August 2020–May 2024

B.S. in Honors Astronomy/Astrophysics

Advisors: Ryan MacDonald, Ted Bergin, Emily Rauscher

Teaching

Graduate TA, ASTRON 104 (Our Exploration of the Solar System)

September–December 2026

Graduate TA, ASTRON 170 (The Dark Side of the Universe)

January–May 2025

Undergraduate Grader, PHYSICS 406 (Statistical and Thermal Physics)

September–December 2023

Academic Service

American Association for the Advancement of Science Mass Media Fellowship

June–August 2026

American Astronomical Society (AAS) Congressional Visit Days Volunteer

April 2025

Skywatching Volunteer at Horwitz-DeRemer Planetarium

October 2025–

UW–Madison Department of Astronomy Telescope Lead

September 2025–

Washburn Observatory Volunteer

September 2024–

Honors and Awards

NASA Wisconsin Space Grant Consortium: Graduate & Professional Research Fellowship

Summer 2026

Grant award: **\$4,381**

Blair Savage Astronomy Travel Fund

Winter 2026

Grant award: **\$750**

University Honors/Dean's List/James B. Angell Scholar

Fall 2020–Winter 2024

Michigan Competitive Scholarship

Fall 2020–Winter 2023

Frank J. Shay Scholarship

Fall 2022–Winter 2023

Regents Merit Scholarship

Fall 2020–Winter 2021

Selected Works

[11] “Chemical Divergence and Water Depletion: Gas Properties of Evolved Upper Scorpius Disks Revealed by JWST/MIRI”

Raul, E., et al., 2026, manuscript submitted to AJ

[10] “From Young to Older Disks: JWST/MIRI Evidence for Fading Molecular Emission and Hints for Elevated C/O in Upper Scorpius”

Chengyan, Xie., et al. (including **Raul, E.**), manuscript accepted to AJ

[9] “JWST/MIRI Hydrocarbon and Water Absorption in the Wind of a Young Disk: Signatures of Pebble Drift and Carbon Grain Sublimation”

Colmenares, MJ., et al. (including **Raul, E.**), 2026, ApJ 1002 176

[8] “Possible presence of volatiles on the warm super-Earth TOI-270 b”

Coulombe, LP., et al. (including **Raul, E.**), 2025, AJ, 170, 226

[7] “The JDISC Survey: Linking the Physics and Chemistry of Inner and Outer Protoplanetary Disk Zones”

Arulanantham, N., et al. (including **Raul, E.**), 2025, AJ, 170, 67

[6] “Escaping Helium, Depleted Methane, and Hints of Carbon Dioxide Suggest a High-Metallicity Atmosphere on Sub-Neptune GJ 3090 b from JWST Transit Spectroscopy”

Ahrer, E., et al. (including **Raul, E.**), 2025, ApJL, 985, L10

[5] “Tracking the Evolution of Hydrocarbons Through Carbon Grain Destruction in Protoplanetary Disks”

Raul, E., Alarcón, F., Bergin, E. A, 2025, ApJ, 982, 155

[4] “JWST/NIRISS Reveals the Water-rich "Steam World" Atmosphere of GJ 9827 d”

Piaulet-Ghorayeb, C., et al. (including **Raul, E.**), 2024, ApJL, 974, L10

[3] “JWST Reveals CH₄, CO₂, and H₂O in a Metal-rich Miscible Atmosphere on a Two-Earth-Radius Exoplanet”

Benneke, B., et al. (including **Raul, E.**), 2024, manuscript submitted to ApJ

[2] “Recent Swift Observations of NGC 5907 ULX-1”

Raul, E., Bacon, S., Brumback, M., Hoin, S., Miller, J. M., Reynolds, M. T., 2022, ATEL 15809

[1] “Swift Observations of the M51 ULX RX J133007+47110”

Raul, E., Brumback, M., Miller, J. M., Reynolds, M. T., et al., 2022, ATEL 15803

Selected Presentations

Contributed Talk, Discs on the Exe, Exeter, UK July 2026

Contributed Talk, Emerging Researchers in Exoplanet Science Symposium XI, Ann Arbor, MI July 2026

Contributed Talk, Cool Stars 23, Tokyo, Japan June 2026

Contributed Talk/**SOC**, Great Lakes Exoplanet Area Meeting, Madison, WI November 2025

Contributed Talk/Dwornik Judge, Lunar & Planetary Science Conference, The Woodlands, TX March 2025

Contributed Talk, Bay Area Exoplanet Meeting 47, San Francisco, CA December 2024

Contributed Poster, Astrobiology and the Future of Life Meeting, Houston, TX October 2024

Contributed Talk, Emerging Researchers in Exoplanet Science Symposium IX, Ithaca, NY July 2024

Contributed Talk, American Astronomical Society Meeting 243, New Orleans, LA January 2024

Contributed Lightning Talk, Great Lakes Exoplanet Area Meeting, Bloomington, IN October 2023

Public Talks and Outreach

Organizer of Stargazing at Science on the Square (Wisconsin Science Festival)	October 2026
Talk at Yellowstone Lake State Park, WI	August 2026
Talk at Wyalusing State Park, WI	August 2026
Talk at Kettle Moraine State Park, WI	May 2026
Graduate Research Showcase at the Capitol, WI	April 2026
Science Night Presenter at Emerson Elementary School, Madison, WI	March 2026
Talk at Eisenhower Middle-High School, New Berlin, WI	January 2026
1st Grade Astronomy Unit Teacher at Van Hise Elementary School, Madison, WI	January 2026
Organizer of Stargazing at Science on the Square (Wisconsin Science Festival)	October 2025
Talk at Buckhorn State Park, WI	August 2025
Talk at Newport State Park, WI	August 2025
Talk at Monona Public Library, WI	August 2025
Science Volunteer at UW–Madison Wonders of Physics Fair	February 2025
Physics Demonstrations Volunteer at Metro Detroit Youth Clubs	November 2023

Science Communication and Press Releases

MIT Technology Review - “Companies are starting to ask AI what you think instead of you”	October 2026
MIT Technology Review - “Fervo Energy and its plan to streamline geothermal power”	October 2026
MIT Technology Review - “35 Innovators Under 35 2026: Shuang Li”	September 2026
MIT Technology Review - “35 Innovators Under 35 2026: Hannah Earley”	September 2026
MIT Technology Review - “This founder is teaching chips how to recycle (their energy)”	September 2026
MIT Technology Review - “How to sign up for a virtual power plant—and decide whether you should”	August 2026
MIT Technology Review - “Shape-shifting mirrors on NASA’s new space telescope could unveil Jupiters like our own”	July 2026
AAS - “Rocket Science Needs Rocket Fuel: An Early Scientist’s Experience on the Hill”	May 2025
Universe Today Podcast - “First Ever Steam World Found by JWST”	December 2024
University of Michigan - “JWST finds first exotic 'steam world' shrouded in water vapor”	October 2024

Accepted Observing Proposals

JWST Cycle 5, 57.6 hours, GO 12157 (PI: Benneke, B.)	2026
MIRI Clarity: Uncovering the Mysteries of Water Worlds and Sub-Neptunes in the Mid-Infrared	
JWST Cycle 4, 68.5 hours, GO 7135 (PI: Zhang, K.)	2025
Population Study of Chemistry in the First Million Years of Planet Formation: A MIRI Survey of Embedded Disks in Ophiuchus	

Science Teams

ACES: Architecture of Compact Exoplanetary Systems (PI: Becker J.)	2025-2026
JDISCS: JWST Disk Infrared Spectral Chemistry Survey (PIs: Salyk C., Banzatti A., Pontoppidan K.)	2024-2026
JWST Water Worlds Survey Team (Member, PI: Benneke B.)	2024-